

EB-JEPA for Clinical EEG

Self-Supervised Abnormality Detection

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Why Self-Supervised EEG?

Clinical reality:

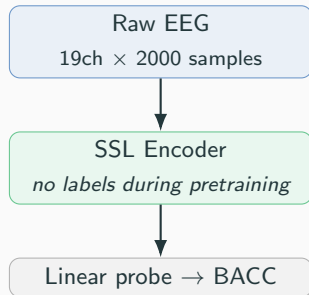
- Every EEG must be reviewed by a licensed neurologist
- Labels are scarce and expensive
- Unlabelled EEG data is abundant

Goal: learn a *transferable* EEG representation **without labels**

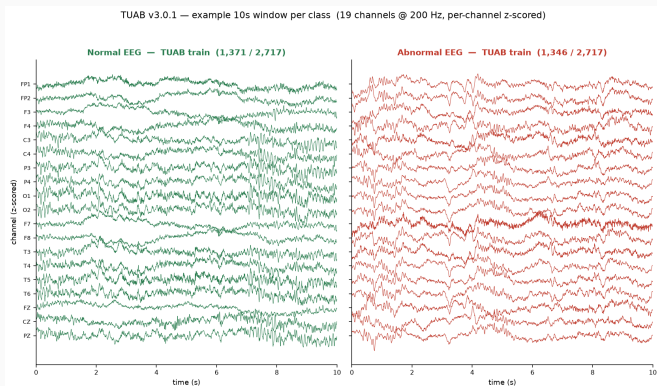
Hypothesis: corruption augmentation + weak invariance ($\lambda=1$) is the dominant lever, not encoder scale.

Benchmark: TUAB v3.0.1

- Binary: normal / abnormal clinical EEG
- 19 channels, 200 Hz, 10 s windows
- 2717 train / 276 eval (patient-disjoint)



What the Data Looks Like



One 10 s window per class, 19 channels @ 200 Hz, per-channel z-scored — exactly what the encoder sees, no spectrogram, no hand-crafted features.

What jumps out: the abnormal trace (right) shows visibly higher-amplitude, less rhythmic bursts in several channels around 4–6 s, vs. the normal trace's (left) more uniform, lower-amplitude background rhythm.

This is exactly the kind of structure the SSL encoder must learn to separate – without ever seeing the normal/abnormal label.

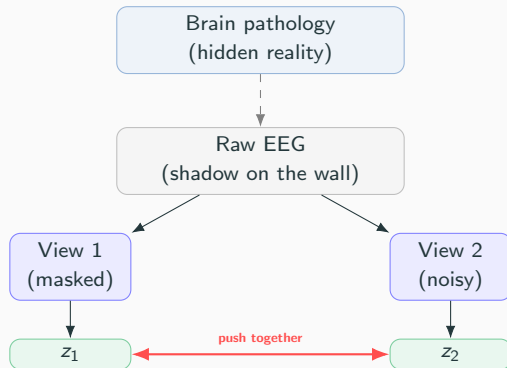
Plato's Cave: What SSL Actually Does

Prisoners in Plato's cave see only **shadows on the wall**, not the real objects. They learn to *predict* patterns in the shadows without ever seeing the cause.

In our setting:

- Cave wall = raw EEG voltages
- Hidden reality = underlying brain pathology
- SSL encoder = prisoner learning that *two corrupted views of the same shadow are the same object*
- Linear probe = asking the prisoner to name what they see

The encoder never sees a neurologist's label. It learns structure from the signal alone.



EB-JEPA: Two-View Invariance + Corruption

Two-view SSL pipeline:

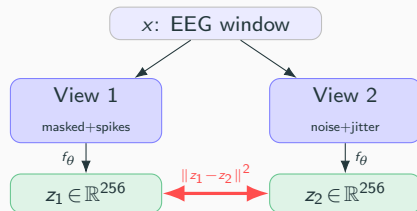
1. Take one EEG window; create two views v_1, v_2
2. Encode: $z_i = f_\theta(v_i) \in \mathbb{R}^{256}$
3. Minimise $\|z_1 - z_2\|^2 + \text{anti-collapse term}$

Standard augmentations: noise, jitter, channel drop

Our design: Corruption augmentation

- 20 % time steps **masked** to zero
- 20 % time steps injected with $\pm 6\sigma$ **spikes**
- Mimics real EEG artefacts; forces robust learning

Objectives: VICReg ($V+C$) and SIGReg (Epps-Pulley)



Finding 1: Corruption Augmentation as Dominant Lever

Variant	BACC _{REC}	Δ
Base VICReg (no corruption)	0.796	–
+ Corruption (3-seed)	0.819$\pm$0.004	+0.023
+ spectral (best run)	0.836	+0.040
+ spectral + stylised-facts	0.823 $\pm$ 0.007	≈ 0
1.35M conv (3-seed)	0.812 $\pm$ 0.021	–0.007
25% pretraining data	0.816	–0.008

3-seed robust avg: 0.819 ± 0.004 (corruption only); single best run w/ spectral loss: 0.836.

Why corruption helps:

- Standard augmentation creates views that are “too easy” to match
- Masking + spike injection forces learning *content*, not artefact patterns
- Mirrors real EEG noise distribution

What does NOT help:

- Scaling 0.4M $\rightarrow$ 1.35M params $\rightarrow$ same plateau
- 4 $\times$ less pretraining data costs only 0.8 pp $\rightarrow$ not data-starved
- Piling stylised-facts terms on top of spectral $\rightarrow$ flat

Finding 2 (Novel): EEG Prefers Weak VICReg Invariance

VICReg loss:

$$\mathcal{L} = \lambda_{\text{inv}} \underbrace{\|z_1 - z_2\|^2}_{\text{inv}} + 25 V + 1 C$$

Paper-correct: $\lambda_{\text{inv}} = 25$. Default: $\lambda_{\text{inv}} = 1$.

λ_{inv}	BACC _{REC}	Seeds
1 (default)	0.819 $\pm$.004	3
5	0.820	1
10	0.814	1
25 (paper)	0.789 $\pm$.005	3

Gap $\lambda=1$ vs $\lambda=25$: 3 pp, $\approx 6\sigma$, not noise

Why?

- With $\lambda=1$: anti-collapse ($V+C$) **dominates**
- Encoder spreads representations: more discriminative
- With $\lambda=25$: encoder collapses toward view-similarity

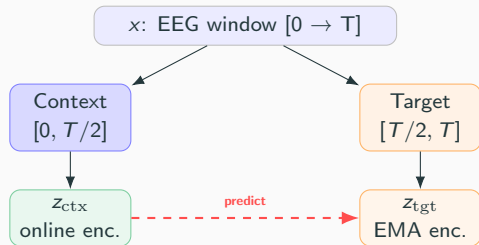
Sweep $\lambda \in \{1, 5, 10, 25\}$: Sweet spot at $\{1, 5\}$; gap widens monotonically.

First evidence that EEG SSL requires weaker view-invariance than vision SSL. Confirmed on a second, independent 1.35M+spectral config: 1(0.824)>5(0.820)>10(0.814)>25(0.804), same monotone order.

NaiveJEPA: Temporal-Split Prediction, No Anti-Collapse Needed

Idea: split one window in time instead of augmenting.

- Context ($0 \rightarrow T/2$): online encoder $\rightarrow$ mean-pool
- Target ($T/2 \rightarrow T$): EMA encoder $\rightarrow$ mean-pool
- Loss: $\text{MSE}(\hat{z}_{\text{tgt}}, z_{\text{tgt}})$
- **No VICReg needed:** collapse resisted by BatchNorm (zero variance $\rightarrow$ explosion) + EMA lag (non-stationary target)



0.04 BACC below VICReg+corruption, with 1 hyperparameter.

Method	BACC _{REC}	Hypers
VICReg + corruption	0.819 ± .004	4
NaiveJEPA	0.779	1

Encoder Ranking: Simple Conv Wins

Validated, bug-free (capacity-matched, BACC):

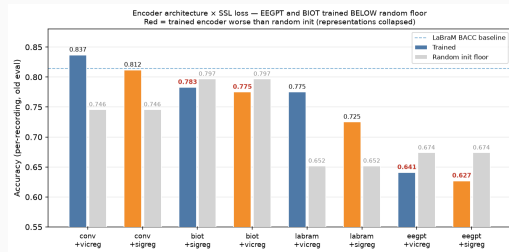
Encoder	Type	Params	BACC _{REC}
Conv1D	Strided	1.35M	0.824
Transformer	Scratch	1.29M	0.725 (−10pp)

Early exploration (pre-bugfix VICReg coeffs, raw accuracy not BACC) had also flagged

LaBraM-/BIOT-style patch transformers (~0.78–0.84 acc) and EEGPT (below-random-floor collapse) as weaker than conv — directionally consistent with the validated result above, but not re-measured post-fix.

EEGPT failure (confirmed, post-fix):

cross-attention pools 19 ch → 1 token;
channel-drop creates incompatible views. Fix:
remove channel-drop → TUEV 0.182 → 0.358.



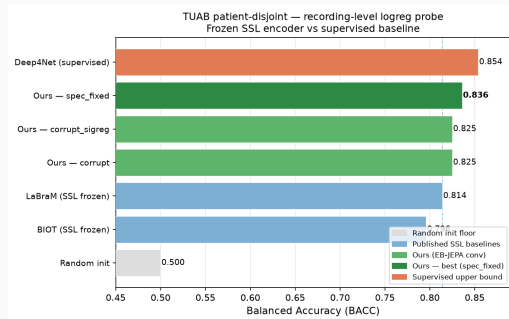
Position vs Literature

Evaluation trap: literature (BIOT, LaBraM) reports **per-window** BACC. Our setup mean-pools 16 windows/patient → **per-recording** (+5 pp for same encoder). Direct comparison manufactures a false win.

Method	BACC _{WIN}	AUROC
ContraWR (SSL)	0.775	0.846
BIOT pretrained	0.802	0.874
LaBraM-Base	0.814	0.902
EB-JEPA SIGReg+corrupt	0.775	0.856
EB-JEPA VICReg+corrupt	0.770	0.848
EEGNet (supervised)	0.796	0.882

✓ Matches ContraWR (1.6M) with 0.4M params

✗ 3.9 pp below LaBraM (5.8M, 200 ep)



Cross-Task Transfer to TUEV (6-class Event Detection)

Protocol: freeze TUAB-pretrained encoder; train linear probe on TUEV (zero TUEV labels during SSL).

Method	TUEV BACC	Labels?
ST-Transformer	0.3385	full train
CNN-Transformer	0.4100	full train
VICReg+corrupt (TUAB SSL)	0.382	probe only
NaiveJEPA (TUAB SSL)	0.389	probe only
+spectral (best)	0.425	probe only
FFCL	0.4551	full train
ContraWR	0.4664	full train
BIOT pretrained	0.5009	full train

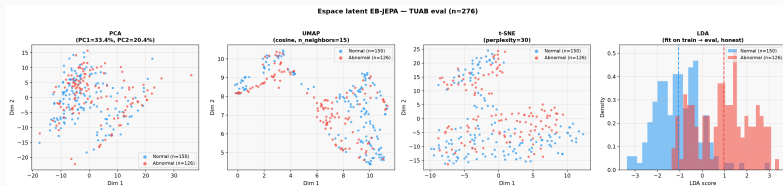
Zero TUEV labels seen during SSL pretraining. A linear probe on frozen features beats **ST-Transformer** and **CNN-Transformer** trained end-to-end on TUEV.

Transfer from a binary task to a 6-class task with only logistic regression: strongest evidence of genuine generality.

We trail BIOT/ContraWR/FFCL because they fine-tune or train from scratch on TUEV.

TUEV is severely imbalanced: BCKG is 64% of events, SPSW only 0.77%. A BCKG-only predictor scores 64% accuracy while failing the task — why we report BACC, not accuracy.

Latent Space: EB-JEPA Representations



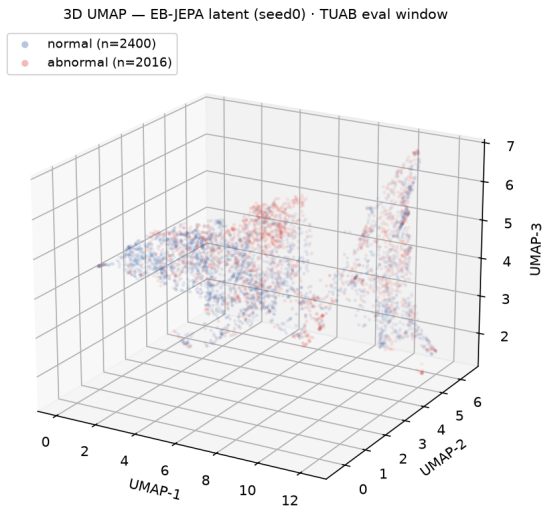
Setup: Conv1D +
VICReg $\lambda=1$ +
corruption (BACC
0.819). 276 eval
patients, 256-dim.

PCA/UMAP/t-SNE
look mixed. VICReg
spreads signal across all
256 dims equally; 2D
projections lose the
discriminative direction.
Expected, not a failure.

LDA (4th): fit on *train*
(2717), projected on
eval (honest).

Near-clean separation
confirms BACC 0.819 is

Latent Space: 3D UMAP (EB-JEPA, Window-Level)



Setup: same encoder, window-level latents (seed 0), 3D UMAP instead of 2D.

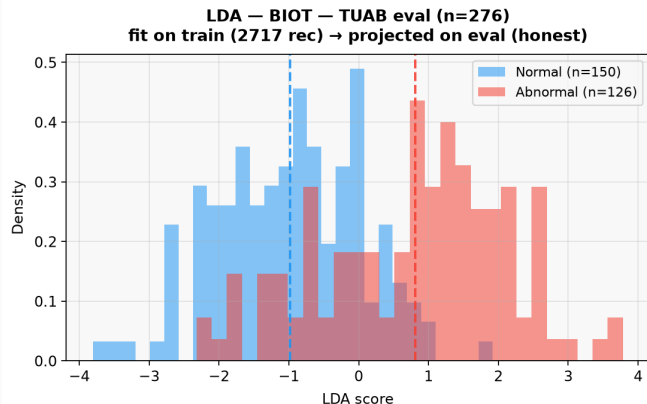
2400 normal / 2016 abnormal windows. The third dimension recovers a two-lobe structure that 2D UMAP collapses.

Still unsupervised projection (no labels used to fit UMAP) — qualitative support for Finding 1, not the BACC evidence (that's LDA, previous slide).

Latent Space: BIOT (LDA, Same Protocol)

BIOT pretrained: EEG-SHHS+PREST-18ch

BACC_{WIN} 0.802 (sleep+pathology pretraining, 1.2M)



Same LDA protocol as EB-JEPA (slide before last): fit on TUAB *train* (2717 rec), projected on *eval* (276 rec, honest).

EB-JEPA: BACC_{REC} 0.819, 0.4M params.

BIOT pretrained on sleep EEG (SHHS); domain mismatch with clinical pathology may reduce separability.

Summary

What works

- **Corruption:** +2.3 pp BACC (3-seed, robust)
- $\lambda_{\text{inv}}=1$: beats $\lambda=25$ by 3 pp/ 6σ , EEG-specific
- **SIGReg** $\geq$ **VICReg**: consistent +0.5–1 pp AUROC
- **NaiveJEPA**: 0.779 BACC with 1 hyperparameter
- **TUEV transfer**: beats 2 supervised baselines, zero TUEV labels

Open next step

λ_{inv} sweep $\{1, 5, 10, 25\}$ confirms monotone ordering: EEG SSL prefers weaker invariance, first systematic evidence in the literature.

What doesn't

- Scaling the encoder (capacity is not the limit)
- $4\times$ pretraining data (conv saturates)
- EEGPT (channel-drop $\times$ channel-pool conflict)
- Spectral auxiliary loss (neutral)
- GRU predictive JEPA (same-window degenerates)

Thank you

EB-JEPA for Clinical EEG — self-supervised, frozen, label-free pretraining

Questions welcome

0.836 BACC_{REC} (best)
0.819 $\pm$.004 robust 3-seed

Corruption: +2.3 pp
dominant lever

$\lambda_{\text{inv}}=1$ beats 25
EEG-specific finding

Backup slides follow: per-recording table, λ_{inv} ablation.

Backup: Per-Recording Table (Clinical Metric)

Model	BACC _{REC}	AUROC	Eval
Conv + VICReg + spectral 0.1 (best single run)	0.836	0.887	Frozen
Conv + VICReg + corruption (3 seeds)	0.819 $\pm$.004	0.900 $\pm$.006	Frozen
Conv + SIGReg + corruption	0.825	0.913	Frozen
Conv + NaiveJEPA	0.779	0.863	Frozen
Conv + VICReg + corruption (FT, 3s)	0.812 $\pm$.004	0.908 $\pm$.006	Fine-tuned
EEGNet (supervised)	0.812	0.882	–
LaBraM-Base (per-win only)	<i>0.814_{WIN}</i>	<i>0.902_{WIN}</i>	FT

Per-recording = mean-pool 16 windows/patient. Not directly comparable to per-window literature. Best per-window: 0.775 (SIGReg+corrupt).

Backup: VICReg λ_{inv} Ablation

λ_{inv}	BACC _{REC}	AUROC	Note
1 (default)	0.819 $\pm$.004	0.900 $\pm$.006	anti-collapse dominates
5	0.820	0.893	$\approx$ tied with $\lambda=1$
10	0.814	0.900	slight drop
25 (paper)	0.789 $\pm$.005	0.882 $\pm$.004	invariance dominates

Sweet spot $\lambda \in \{1, 5\}$; gap vs $\lambda = 25$: 3 pp, $\approx 6\sigma$.

0.4M corruption-only config above; confirmed on independent 1.35M+spectral config: 1(.824)>5(.820)>10(.814)>25